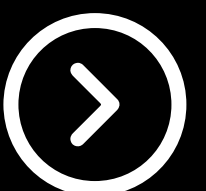


Reinforcement Learning (RL) Cheat Sheet



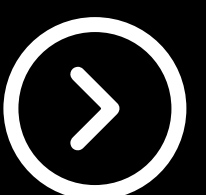
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1. Building Blocks of Reinforcement Learning

RL is an area of machine learning where an agent learns to make decisions by interacting with an environment to maximize a cumulative reward.

Concept	Definition	Analogy
 Agent	The learner or decision-maker	The student or player.
 Environment	The world the agent interacts with. It receives actions and returns new states and rewards.	The classroom, game board, or world.
 State (s)	A complete description of the current situation.	Where the player is on the board.
 Action (a)	A move the agent can make from a given state	Moving a chess piece or choosing an answer
 Policy (π)	The agent's strategy; a mapping from states to actions. It tells the agent what action to take in each state.	The strategy guide or a student's study plan.
 Reward (R)	A scalar feedback signal that indicates how well the agent performed in a single step. The agent's goal is to maximize the cumulative reward.	Points scored or a positive/negative grade.
 Value Function ($V(s)$ or $Q(s,a)$)	A prediction of the expected future cumulative reward from a given state (V) or after taking a specific action from a state (Q).	The total expected payout from a certain position in a game.

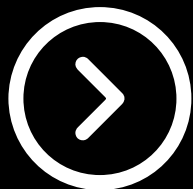


2. Algorithms and Advanced Concepts

These concepts cover methods for learning the optimal policy and advanced techniques for integrating human preferences.

Concept	Description	Type
 Model-Free RL	The agent learns the optimal policy or value function directly from experience without explicitly building a model of the environment's dynamics.	Common examples: Q-Learning, Sarsa, DQN (Deep Q-Network), Policy Gradient methods.
 Policy Gradient	Algorithms that directly optimize the policy (π) by calculating the gradient of the expected reward with respect to the policy parameters.	Examples: REINFORCE, A2C (Advantage Actor-Critic).
 Actor-Critic	Combines two components: the Actor (updates the policy) and the Critic (estimates the value function, helping the actor update more efficiently).	Examples: A2C/A3C, DDPG, TD3, PPO.
 Proximal Policy Optimization (PPO)	A policy gradient method that is one of the most widely used and stable RL algorithms. It restricts the amount the policy can change at each step to ensure stable learning.	The standard for Reinforcement Learning from Human Feedback (RLHF).

Concept	Description	Application in Generative AI
Reinforcement Learning from Human Feedback (RLHF)	A process where an RL agent is trained using a Reward Model (RM) that has been trained on human preference data. The goal is to align the model's output with human values/instructions.	Crucial for making Large Language Models (LLMs) helpful and harmless.
Reward Model (RM)	A model (usually a separate neural network) that takes an LLM's prompt and response as input and outputs a scalar reward (a score) that quantifies how good/preferred the response is.	Replaces the standard environment reward in RLHF.
Direct Preference Optimization (DPO)	An alternative to RLHF that simplifies the alignment process. DPO reframes the optimization problem as a simple classification task on the preference data, directly training the policy to satisfy the preferences without explicitly training a separate Reward Model or using a complex RL algorithm like PPO.	More stable and computationally simpler than PPO-based RLHF.



3. Generative AI Use Cases

RL is primarily used in Generative AI for alignment and fine-tuning to ensure models produce high-quality, safe, and helpful outputs.

1. Reinforcement Learning Fine-Tuning (RLFT)



Goal: To adapt a pre-trained model (like an LLM) to a new objective that is difficult to express with standard loss functions, such as maximizing coherence, truthfulness, or adherence to complex instructions.

RL Components:

Agent: The LLM being fine-tuned (its parameters are the policy).

Action: Generating the next token/word in the sequence.

Environment: The state space (context and previous tokens) and a custom metric or external evaluator that provides the reward.

Reward: A score provided by a metric (e.g., ROUGE, BLEU) or a specially trained Reward Model (RM).

Algorithm

Usually PPO or another Policy Gradient method.

2. Alignment with Human Feedback (RLHF)



Goal: Aligning an LLM's behavior with complex human preferences, safety standards, and instruction-following.

Process Overview (PPO-based RLHF):

- 1. Data Collection:** Humans label pairs of model outputs, indicating which one they prefer.
- 2. Reward Model (RM) Training:** Train a separate model on this human preference data to predict a score (the reward) for any given response.
- 3. RL Fine-Tuning:** Use the trained RM as the reward function to fine-tune the LLM with an RL algorithm (typically PPO). A KL-divergence penalty is often added to the reward to keep the fine-tuned model close to the original base model, preventing it from generating irrelevant text just to chase a high reward.

3. Direct Preference Optimization (DPO)

Goal: Achieving the same alignment outcome as RLHF but with a simpler, direct training method.

Mechanism: Instead of training a separate RM and then using RL, DPO directly optimizes the policy (π) to prefer the 'chosen' response over the 'rejected' response based on the human preference data.

RL Component Connection: While it doesn't use a traditional RL algorithm, DPO's loss function is derived from the optimal policy objective in RL (specifically, in the context of a Reward Model), effectively achieving the optimal policy that satisfies the human preferences.

Advantage: Simplicity and Stability—eliminates the need for a separate RM and the complexities of sampling from an LLM within an RL loop.